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Antibacterial and Antifungal Efficacy of n-Hexane Extract of *Ocimum gratissimum* Against Clinical Pathogens: An In Vitro Assessment

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Abstract

Background: Antimicrobial resistance poses a critical global health challenge, necessitating exploration of alternative therapeutic agents from natural sources. *Ocimum gratissimum* (scent leaf) has documented traditional use against infectious diseases in West African communities.

Objective: To evaluate the antimicrobial activity of n-hexane extract of *O. gratissimum* leaves against selected bacterial and fungal isolates.

Methods: Pre-characterized clinical isolates of *Salmonella typhi*, *Escherichia coli*, *Candida albicans*, and *Trichophyton canis* were cultured. The n-hexane extract was prepared at concentrations of 20%, 40%, and 60% using dimethyl sulfoxide as diluent. Antimicrobial assay employed agar well diffusion method with Mueller-Hinton agar for bacteria and Sabouraud's Dextrose agar for fungi. Ciprofloxacin (500mg) and ketoconazole (200mg) served as positive controls.

Results: Concentration-dependent antibacterial activity was observed. Against *E. coli*, zones of inhibition measured 20mm (60%), 17mm (40%), and 15mm (20%). Against *S. typhi*, measurements were 17mm (60%), 16mm (40%), and 15mm (20%). Antifungal activity was limited: *C. albicans* showed 5mm inhibition at 60% concentration only, while *T. canis* exhibited 6–7mm zones at lower concentrations.

Conclusion: The n-hexane extract demonstrated superior antibacterial compared to antifungal activity, suggesting selective bioactive compound penetration of bacterial cell structures. These findings support the traditional use of *O. gratissimum* in treating bacterial infections and its potential incorporation into antimicrobial drug development.

Keywords: Antimicrobial activity, *Ocimum gratissimum*, bacterial inhibition, fungal resistance, natural antibiotics, agar well diffusion

1. Introduction

The rapid emergence and spread of antimicrobial-resistant pathogens represent a significant threat to global health systems. The overuse and misuse of synthetic antibiotics have accelerated resistance, reducing therapeutic options for infectious diseases (WHO, 2020). Consequently, there is increasing interest in natural products, particularly medicinal plants, as alternative sources of antimicrobial agents.

Ocimum gratissimum Linn., commonly known as scent leaf, belongs to the family Lamiaceae and is extensively cultivated in tropical regions of Africa. The plant is traditionally used for treating diarrhea, respiratory

infections, skin diseases, and wound infections (Sofowora, 2008). Phytochemical studies have reported that *O. gratissimum* contains bioactive compounds such as alkaloids, flavonoids, tannins, saponins, and essential oils, many of which possess antimicrobial properties.

While several studies have investigated aqueous and ethanolic extracts of *O. gratissimum*, limited data exist on the antimicrobial potential of its n-hexane extract, particularly against both bacterial and fungal clinical isolates. This study therefore aimed to evaluate the in vitro antibacterial and antifungal activities of n-hexane leaf extract of *O. gratissimum* against selected clinical pathogens.

2. Materials and Methods

2.1 Plant Material Collection and Identification

Fresh leaves of *Ocimum gratissimum* were collected from a local environment in Sokoto State, Nigeria. The plant was authenticated by a botanist in the Department of Biological Sciences, Usmanu Danfodiyo University, Sokoto. The leaves were washed, air-dried at room temperature, and pulverized into fine powder.

2.2 Preparation of n-Hexane Extract

The powdered plant material was subjected to cold maceration using n-hexane as solvent for 72 hours with intermittent shaking. The extract was filtered and concentrated using a rotary evaporator to obtain a crude n-hexane extract. The extract was

stored in airtight containers at 4°C until use.

2.3 Test Microorganisms

Pre-characterized clinical isolates of *Escherichia coli* and *Salmonella typhi* (bacteria), and *Candida albicans* and *Trichophyton canis* (fungi) were obtained from a medical microbiology laboratory. The organisms were confirmed using standard biochemical and microscopic methods.

2.4 Preparation of Extract Concentrations

The crude extract was reconstituted in dimethyl sulfoxide (DMSO) to obtain concentrations of 20%, 40%, and 60%.

2.5 Antimicrobial Assay

The agar well diffusion method was employed. Mueller–Hinton agar plates were used for bacterial cultures, while Sabouraud Dextrose agar plates were used for fungal cultures. Standardized

inocula were evenly spread on agar surfaces. Wells were bored into the agar and filled with different extract concentrations.

Ciprofloxacin (500 mg) and ketoconazole (200 mg) served as positive controls, while DMSO served as negative control. Plates were incubated at 37°C for

bacteria (24 hours) and 25–28°C for fungi (48–72 hours).

2.6 Measurement of Zones of Inhibition

Zones of inhibition were measured in millimeters using a transparent ruler. All experiments were conducted in triplicate, and mean values were recorded.

3. Results

The n-hexane extract of *O. gratissimum* showed notable antibacterial activity against *E. coli* and *S. typhi*. The highest inhibition

was observed at 60% concentration. Antifungal activity was minimal, with limited inhibition observed only at higher concentrations.

Table 1: Zones of Inhibition (mm)

of n-Hexane Extract of *O.*

gratissimum

Microorganism	20%	40%	60%	Positive Control
<i>E. coli</i>	15	17	20	28
<i>S. typhi</i>	15	16	17	26
<i>C. albicans</i>	—	—	5	24
<i>T. canis</i>	6	7	—	22

4. Discussion

The results indicate that the n-hexane extract of *O. gratissimum* possesses significant antibacterial activity, particularly against Gram-negative bacteria. This may be attributed to the presence of non-polar phytochemicals capable of disrupting bacterial cell membranes. The limited antifungal activity observed may be due to

the complex cell wall structure of fungi, which restricts penetration of bioactive compounds.

The findings align with previous reports that demonstrate stronger antibacterial than antifungal activity of *O. gratissimum* extracts (Akinmoladun et al., 2007). The concentration-dependent inhibition further supports the extract's pharmacological potential.

5. Conclusion

The n-hexane leaf extract of *Ocimum gratissimum* exhibited appreciable antibacterial activity against *E. coli* and *S. typhi* but limited antifungal effects against

C. albicans and *T. canis*. These findings scientifically support the traditional use of *O. gratissimum* in the treatment of bacterial infections and suggest its potential as a source of novel antibacterial agents.

Declaration for JQISST

This research was conducted in accordance with ethical guidelines for scientific inquiry within Islamic frameworks, emphasizing beneficial knowledge (al-'ilm al-nafi'), environmental respect, and service to human welfare. No human or animal subjects were harmed in this study.

Conflict of Interest: The authors declare no conflict of interest.

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